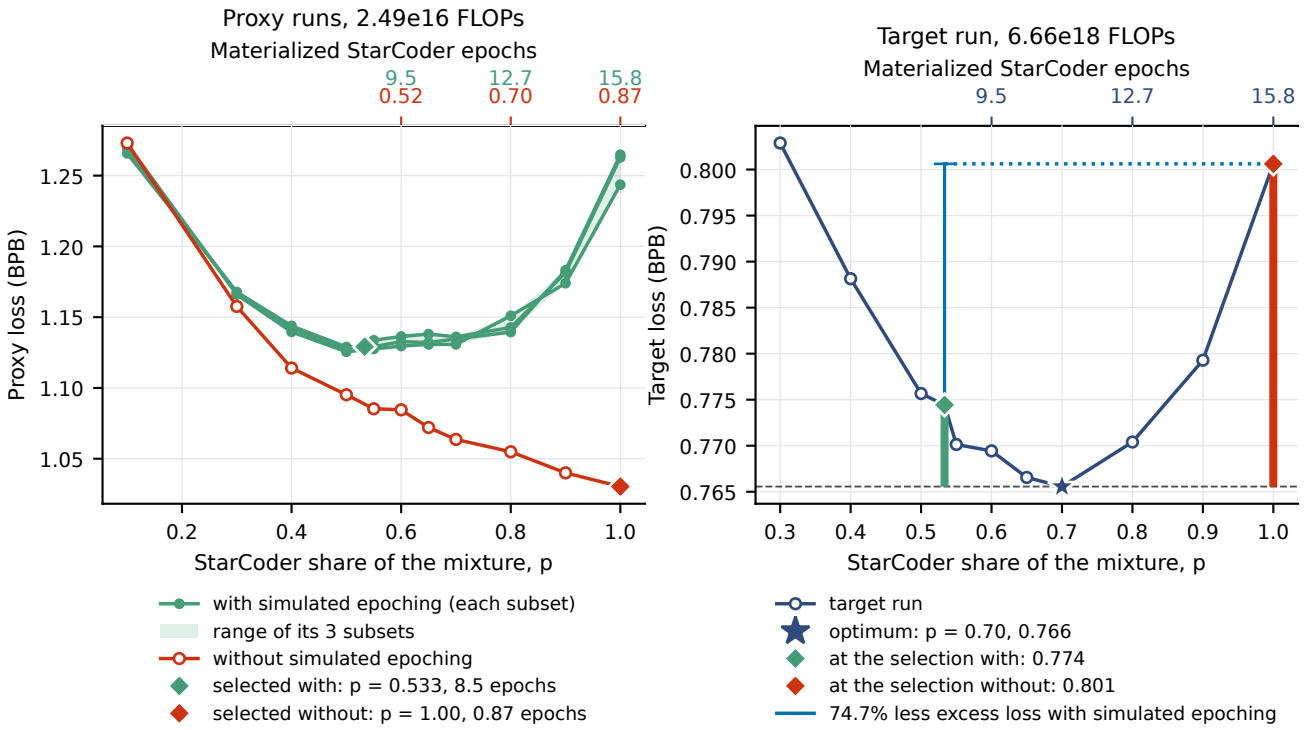
Proxy runs, 2.49e16 FLOPs
Materialized StarCoder epochs
9.5
12.7
15.8
0.52
0.70
0.87
Proxy loss (BPB)
1.25
1.20
1.15
1.10
1.05
0.2
0.4
0.6
0.8
1.0
StarCoder share of the mixture, p
with simulated epoching (each subset)
range of its 3 subsets
without simulated epoching
selected with: p = 0.533, 8.5 epochs
selected without: p = 1.00, 0.87 epochs
Target run, 6.66e18 FLOPs
Materialized StarCoder epochs
9.5
12.7
15.8
Target loss (BPB)
0.800
0.795
0.790
0.785
0.780
0.775
0.770
0.765
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
StarCoder share of the mixture, p
target run
optimum: p = 0.70, 0.766
at the selection with: 0.774
at the selection without: 0.801
74.7% less excess loss with simulated epoching